

# Cambridge Lower Secondary Checkpoint

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**MATHEMATICS****0862/01**

Paper 1

**April 2025**

MARK SCHEME

Maximum Mark: 50

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**Published**

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Markers were instructed to award marks. It does not indicate the details of the discussions that took place at a Markers' meeting before marking began, which would have considered the acceptability of alternative answers.

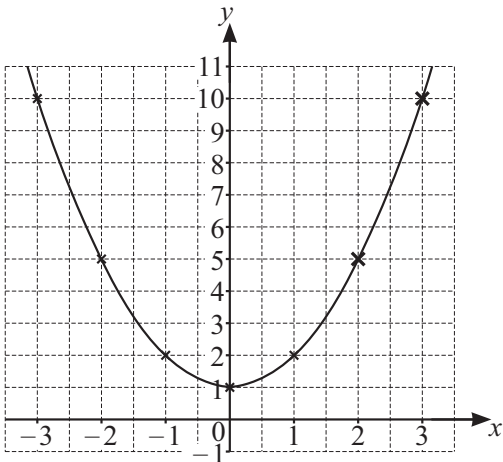
Mark schemes should be read in conjunction with the question paper and the End of Series Report. Cambridge will not enter into discussions about these mark schemes.

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This document has **12** pages.

**Mark scheme annotations and abbreviations**

|           |                            |
|-----------|----------------------------|
| <b>FT</b> | follow through after error |
| <b>SC</b> | special case mark          |
| cao       | correct answer only        |
| dep       | dependent                  |
| isw       | ignore subsequent working  |
| nfw       | not from wrong working     |
| oe        | or equivalent              |
| soi       | seen or implied            |

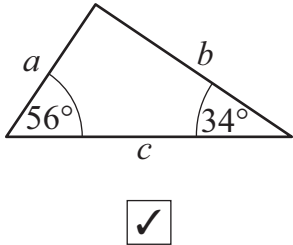
| Question | Answer                                                                                                                                                                                                                                                                                                                                                               | Marks | Part marks                                                                                                                       | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1(a)     | 37                                                                                                                                                                                                                                                                                                                                                                   | 1     |                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1(b)     | $n^2 + 1$                                                                                                                                                                                                                                                                                                                                                            | 1     |                                                                                                                                  | Accept unsimplified, e.g. $n \times n + 1$                                                                                                                                                                                                                                                                                                                                                                                              |
| 2        | <div style="display: flex; justify-content: space-around; align-items: center;"> <div style="text-align: center;">           Quantitative<br/> <input type="checkbox"/><br/> <input checked="" type="checkbox"/> </div> <div style="text-align: center;">           Qualitative<br/> <input checked="" type="checkbox"/><br/> <input type="checkbox"/> </div> </div> | 1     |                                                                                                                                  | Both answers correct for the mark.<br>Accept any clear indication.                                                                                                                                                                                                                                                                                                                                                                      |
| 3        | microlitre                                                                                                                                                                                                                                                                                                                                                           | 1     |                                                                                                                                  | Accept any clear indication.                                                                                                                                                                                                                                                                                                                                                                                                            |
| 4        | $129 = 9t + 11c$                                                                                                                                                                                                                                                                                                                                                     | 1     |                                                                                                                                  | Accept any clear indication.                                                                                                                                                                                                                                                                                                                                                                                                            |
| 5        | $(m =) \sqrt[3]{h}$                                                                                                                                                                                                                                                                                                                                                  | 1     |                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 6        | Correct curve for $-3 \leq x \leq 3$ drawn<br>                                                                                                                                                                                                                                     | 2     | <p>Award 1 mark for 5 and 10 soi</p> <p>or</p> <p>for a correct curve between <math>(-3, 10)</math> and <math>(1, 2)</math>.</p> | <p>For 2 marks, ignore work outside of the region <math>-3 \leq x \leq 3</math></p> <p>For 2 marks, do <b>not</b> accept straight line segments in the region <math>-2 \leq x \leq 2</math></p> <p>Implied by correct points plotted or curve/line through <math>(2, 5)</math> and <math>(3, 10)</math>.<br/>5 and 10 likely to be seen in table.</p> <p>For this mark, ignore work outside of region <math>-3 \leq x \leq 1</math></p> |

| Question | Answer                              |                          |                          | Marks    | Part marks | Guidance                                                           |
|----------|-------------------------------------|--------------------------|--------------------------|----------|------------|--------------------------------------------------------------------|
| 7        | Congruent                           | Not congruent            | Cannot tell              | <b>1</b> |            | Both answers correct for the mark.<br>Accept any clear indication. |
|          | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |          |            |                                                                    |
|          | <input checked="" type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |          |            |                                                                    |

| Question | Answer            | Marks | Part marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------|-------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8        | $\frac{4}{5}$ cao | 3     | <p>Award 2 marks for any of these</p> <ul style="list-style-type: none"> <li><math>\frac{12}{10} \div \frac{15}{10}</math> oe common denominator</li> <li><math>\frac{6}{5} \times \frac{2}{3}</math></li> <li>an answer equivalent to <math>\frac{4}{5}</math></li> </ul> <p><b>or</b></p> <p>Award 1 mark for any of these</p> <ul style="list-style-type: none"> <li><math>\frac{6}{5}</math> <b>and</b> <math>\frac{3}{2}</math> seen</li> <li>1.2 <b>and</b> 1.5 seen</li> <li><math>1\frac{1}{5} \div 1\frac{1}{2}</math> oe or better</li> <li><math>\left(1 + \frac{1}{5}\right) \times \frac{2}{3}</math></li> <li><i>their</i> <math>\frac{6}{5} \times \frac{2}{3}</math> or better oe</li> <li><math>\frac{6}{5} \times</math> the reciprocal of <i>their</i> <math>\frac{3}{2}</math> oe.</li> </ul> | <p>Accept <math>12 \div 15</math> for 2 marks but <math>1.2 \div 1.5</math> is not far enough for 2 marks.</p> <p>e.g. 0.8, must be integer numerator and denominator for the fraction, e.g. <math>\frac{12}{15}</math></p> <p>Or better, e.g. <math>1\frac{1}{5} \div \frac{3}{2}</math></p> <p>oe, e.g. the correct evaluation of this.</p> <p>oe, e.g. the correct evaluation of this <b>or</b> dividing fractions with common denom. Do <b>not</b> accept if either the numerator or denominator of <i>their</i> <math>\frac{3}{2}</math> is = 0 or if <i>their</i> numerator = <i>their</i> denominator.</p> |

| Question       | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks | Part marks                            | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|----|---|----|---|----|---|----------------|----|-----|-----|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| 9              | 8 and 9                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |                                       | Both answers correct for the mark.<br>Do <b>not</b> accept $\sqrt{64}$ and $\sqrt{81}$                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| 10             | <p>A correct reason about sample size <b>and</b> a correct reason about age range.</p> <p><b>Sample size</b>, e.g.</p> <ul style="list-style-type: none"><li>• Her sample size is small</li><li>• She has only chosen 5 students</li><li>• She does not have enough data.</li></ul> <p><b>Age range</b>, e.g.</p> <ul style="list-style-type: none"><li>• She chooses students who are the same age</li><li>• She only chose 18-year-olds</li><li>• 18-year-old students are likely to be taller (than the average for the school)</li><li>• She hasn't picked other ages</li><li>• She has only chosen older students</li><li>• She hasn't picked a mix of classes</li><li>• She has only chosen one class.</li></ul> | 2     | Award 1 mark for each correct reason. | <p>2 marks may be awarded in one answer space.</p> <p>Do <b>not</b> accept two answer spaces saying the same point in different ways for 2 marks.</p> <p>Ignore non-contradictory statements.</p> <p>Do <b>not</b> accept unrelated suggestions as to what she should do instead/incorrect suggestions of non-randomness, e.g.</p> <ul style="list-style-type: none"><li>• She should choose equal numbers of boys and girls</li><li>• She should choose (5) students of different heights rather than random students.</li></ul> |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| 11             | <p>Three different correct pairs of values for <math>a</math> and <math>b</math>, e.g.</p> <table><tr><th><math>a</math></th><th><math>b</math></th></tr><tr><td>4</td><td>2</td></tr><tr><td>2</td><td>4</td></tr><tr><td>16</td><td>1</td></tr><tr><td>−4</td><td>2</td></tr><tr><td>−2</td><td>4</td></tr><tr><td><math>\frac{1}{16}</math></td><td>−1</td></tr><tr><td>256</td><td>0.5</td></tr></table>                                                                                                                                                                                                                                                                                                           | $a$   | $b$                                   | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2 | 2 | 4 | 16 | 1 | −4 | 2 | −2 | 4 | $\frac{1}{16}$ | −1 | 256 | 0.5 | 3 | <p>Award 2 marks for <b>two</b> different correct pairs of values for <math>a</math> and <math>b</math>, ignoring errors, omissions or repeats</p> <p><b>or</b></p> <p>Award 1 mark for <b>one</b> correct pair of values for <math>a</math> and <math>b</math>, ignoring errors, omissions or repeats.</p> | Accept any correct pairs of values for $a$ and $b$ . |
| $a$            | $b$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| 4              | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| 2              | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| 16             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| −4             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| −2             | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| $\frac{1}{16}$ | −1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |
| 256            | 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |    |   |    |   |    |   |                |    |     |     |   |                                                                                                                                                                                                                                                                                                             |                                                      |

| Question | Answer                                                                                                                                                                                        | Marks | Part marks                                                                                                                                                                                                                                      | Guidance                                                                                                                                                          |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12(a)    | <div>Correct</div> <div>Not correct</div> <div> <input type="checkbox"/> <input checked="" type="checkbox"/> </div> <div> <input checked="" type="checkbox"/> <input type="checkbox"/> </div> | 1     |                                                                                                                                                                                                                                                 | Both answers correct for the mark.<br>Accept any clear indication.                                                                                                |
| 12(b)    | 20                                                                                                                                                                                            | 1     |                                                                                                                                                                                                                                                 | Accept any clear indication.                                                                                                                                      |
| 13(a)    | Correct triangle drawn at<br>(4, -4), (6, -3), (8, -6)                                                                                                                                        | 2     | Award 1 mark for correct reflection to<br>triangle M (4, -1), (6, 0), (8, -3)<br><br><b>or</b><br><br>for correct translation by $\begin{pmatrix} 0 \\ -3 \end{pmatrix}$ of <i>their</i><br>triangle M.                                         | Both triangles must be on the provided grid<br>for this mark and at least one must be<br>labelled for the FT.<br>Do <b>not</b> award this for a translation of L. |
| 13(b)    | a reflection                                                                                                                                                                                  | 1     |                                                                                                                                                                                                                                                 | Accept any clear indication.                                                                                                                                      |
| 14       | 78                                                                                                                                                                                            | 2     | Award 1 mark for angle 35 <b>or</b> 137<br>correctly marked on diagram<br><b>or</b><br>for correctly showing an angle alternate to<br>43 on the diagram<br><b>or</b><br>for correct calculation leading to the<br>answer, e.g. $180 - 145 + 43$ | e.g. $540 - 90 - 90 - 145 - (180 - 43)$ ,<br>$360 - 145 - 137$                                                                                                    |
| 15       | 81                                                                                                                                                                                            | 2     | Award 1 mark for (numerator =) 15<br><b>or</b> $11 - 8$ or better.                                                                                                                                                                              | Or better, e.g. $\frac{15}{5}$ , $3^4$ but <b>not</b> just 3 alone.                                                                                               |

| Question | Answer                                                                                                                              | Marks | Part marks                                                                                                                                                                                                                 | Guidance                                                                                                                                                                                                                                                                                                                        |
|----------|-------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16       | $(x =) 6$<br>$(y =) -5$                                                                                                             | 3     | Award 2 marks for $x = 6$ <b>or</b> $y = -5$<br><br><b>or</b><br><br>Award 1 mark for correct method to eliminate a variable<br><br><b>or</b><br><br>for <i>their</i> two values satisfying one of the original equations. | Correct method, e.g.<br>Multiplying one or both equations by a constant to get equal coefficients and correct consistent addition or subtraction<br><b>or</b> making $x$ or $y$ the subject and substituting into the other equation.<br>Implied by, e.g. $-7y = 35$ , $7x = 42$<br>$2(21 + 3y) + y = 7$ , $x - 3(7 - 2x) = 21$ |
| 17       | 61                                                                                                                                  | 2     | Award 1 mark for 25                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                 |
| 18       | A negative integer $n$ and a positive integer $p$ such that $n + p < 0$<br>e.g.<br>$(n =) -3$ , $(p =) 2$<br>$(n =) -5$ , $(p =) 1$ | 1     |                                                                                                                                                                                                                            | Do <b>not</b> accept, e.g. $n = (-3)^2$ , $p = 2^2$                                                                                                                                                                                                                                                                             |
| 19       |                                                  | 1     |                                                                                                                                                                                                                            | Accept any clear indication.                                                                                                                                                                                                                                                                                                    |

| Question | Answer                                                      | Marks | Part marks                                                                                                                                                                                                                                                                       | Guidance                                                                                                                                                                                                                                                                                                                  |
|----------|-------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 20       | Finding the number of pencils is 46 with supporting working | 3     | <p>Award 2 marks for <math>11 + 3 = 7x - 5x</math> or better</p> <p><b>or</b></p> <p>Award 1 mark for <math>5x + 11 = 7x - 3</math> oe</p> <p><b>and</b></p> <p>Award 1 mark for <math>5 \times \text{their } 7 + 11</math><br/>or <math>7 \times \text{their } 7 - 3</math></p> | <p>Minimum working is 46 found <b>and</b> either <math>x = 7</math> or <math>5 \times 7 + 11</math> or <math>7 \times 7 - 3</math></p> <p>Or better, e.g. <math>14 = 2x</math>, <math>x = 7</math></p> <p>Accept with all signs reversed.</p> <p>Award 2 marks for answer 46 with no/insufficient supporting working.</p> |
| 21(a)    | independent                                                 | 1     |                                                                                                                                                                                                                                                                                  | Accept any clear indication.                                                                                                                                                                                                                                                                                              |
| 21(b)    | 40                                                          | 1     |                                                                                                                                                                                                                                                                                  | Accept '40 out of 140' but not $\frac{40}{140}$                                                                                                                                                                                                                                                                           |

| Question | Answer                                                                                                 | Marks | Part marks                                                                                                                                                                                             | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
|----------|--------------------------------------------------------------------------------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|--|---|--|---|--|---|---|--|--|--|--|--|--|--|---|---|--|---|--|---|--|---|---|--|--|--|--|--|--|--|---|---|--|---|--|---|--|---|---|--|--|--|--|--|--|--|---|---|--|---|--|---|--|---|
| 21(c)    | $\frac{16}{49}$ oe                                                                                     | 2     | Award 1 mark for $\frac{4}{7} \times \frac{4}{7}$<br><br><b>or</b><br>for answer $\frac{16}{k}$ or $\frac{c}{49}$ where $k > 16, c < 49$<br><br><b>or</b><br>for a fully correct sample space diagram. | Accept equivalent fractions, decimals and percentages correct to at least 2dp, 0.3265..<br><br>Note: confusing odd and even gives answer $\frac{9}{49}$ which scores 1 mark for $\frac{c}{49}$<br>Award mark before any cancelling.<br><br>Sample space diagram, e.g.<br><table><tr><td></td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>1</td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>2</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>3</td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>4</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>5</td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td></tr><tr><td>6</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>7</td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td></tr></table> |   | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 1 | ✓ |  | ✓ |  | ✓ |  | ✓ | 2 |  |  |  |  |  |  |  | 3 | ✓ |  | ✓ |  | ✓ |  | ✓ | 4 |  |  |  |  |  |  |  | 5 | ✓ |  | ✓ |  | ✓ |  | ✓ | 6 |  |  |  |  |  |  |  | 7 | ✓ |  | ✓ |  | ✓ |  | ✓ |
|          | 1                                                                                                      | 2     | 3                                                                                                                                                                                                      | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5 | 6 | 7 |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 1        | ✓                                                                                                      |       | ✓                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ✓ |   | ✓ |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 2        |                                                                                                        |       |                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 3        | ✓                                                                                                      |       | ✓                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ✓ |   | ✓ |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 4        |                                                                                                        |       |                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 5        | ✓                                                                                                      |       | ✓                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ✓ |   | ✓ |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 6        |                                                                                                        |       |                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 7        | ✓                                                                                                      |       | ✓                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | ✓ |   | ✓ |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 22       | Any fraction with an odd denominator that is not equivalent to a recurring decimal, e.g. $\frac{1}{5}$ | 1     |                                                                                                                                                                                                        | Accept, e.g. $\frac{6}{3}, \frac{20}{1}, \frac{5}{5}$<br>isw here and mark original fraction and ignore any evaluation (correct or incorrect)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 23       | (x =) 11<br>(y =) 8                                                                                    | 2     | Award 1 mark for each correct answer.                                                                                                                                                                  | Award 1 mark only for (x =) 19 <sup>11</sup> <b>and</b> (y =) 7 <sup>8</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |
| 24       | 6                                                                                                      | 1     |                                                                                                                                                                                                        | Accept –6<br>Do <b>not</b> accept $\sqrt{36}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |   |   |   |   |   |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |   |  |  |  |  |  |  |  |   |   |  |   |  |   |  |   |

| Question | Answer   | Marks | Part marks                                                                                                                                                                                                                                                                                                                           | Guidance                                                                                                                                                                                                                                    |
|----------|----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 25       | $2w + 3$ | 2     | <p>Award 1 mark for any of these</p> <ul style="list-style-type: none"> <li>• <math>\frac{3t(2w+3)}{3t}</math> or better</li> <li>• <math>\frac{2tw+3t}{t}</math> or better</li> <li>• <math>\frac{6w+9}{3}</math> or better</li> <li>• <math>2w</math> or <math>3</math> correct in a two-term expression as the answer.</li> </ul> | <p>Or better, e.g. <math>\frac{2w+3}{1}</math></p> <p>Do <b>not</b> accept <math>3t\left(\frac{2w+3}{1}\right)</math> alone as better.</p> <p>Allow these as separate fractions.</p> <p>e.g. <math>2w + 9t</math>, <math>6tw + 3</math></p> |

| Question | Answer                  | Marks | Part marks                                                                                                                                                                                                                                                                                                                                  | Guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----------|-------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26       | (x =) 0.7<br>(y =) 0.55 | 3     | <p>Award 2 marks for answer 0.7 <b>or</b> 0.55</p> <p><b>or</b></p> <p>Award 1 mark for any of these</p> <ul style="list-style-type: none"> <li><math>\frac{0.084}{0.12}</math> oe</li> <li><math>\frac{0.084 - 0.12 \times 0.15}{0.12}</math> oe</li> </ul> <p><b>or</b></p> <ul style="list-style-type: none"> <li>0.018 seen.</li> </ul> | <p>Ignore inclusion of unit m with answers.</p> <p>May be implied by <math>\frac{8.4}{12}</math>, <math>\frac{4.2}{6}</math></p> <p>May be implied by <math>\frac{0.084 - 0.018}{0.12}</math>, <math>\frac{0.066}{0.12}</math>, <math>\frac{0.084}{0.12} - 0.15</math></p> <p>For 1 mark, for the first two bullet points accept equivalent calculations in cm<sup>2</sup> and cm, e.g.</p> <p>may be implied by <math>\frac{840}{12}</math>, 70 cm, 55 cm.</p> |